CSE7000 - INTERNSHIP

A Presentation on

TITLE OF THE INTERNSHIP

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Presidency School of Computer Science and Engineering

Name of the Program: **B.Tech(CSE)**
Name of the HoD: **Dr.Nagaraja S R**
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About Company or Organization

- NVIDIA Corporation Founded in 1993, NVIDIA is a leading American technology company headquartered in Santa Clara, California. It designs GPUs and AI computing platforms that power gaming, data centers, autonomous vehicles, and AI research worldwide.
- The company specializes in GPU Design, Accelerated Computing, Artificial Intelligence, Data Centers, Autonomous Vehicles, and the CUDA software platform. Mission: To be the engine of the new industrial revolution, powering the world's AI infrastructure.

Working domain or the technology

Working Domain

- Artificial Intelligence (AI), Computer Vision, Deep Learning, and Construction Safety.**APEX PORTAL** – Construction Safety focuses on using AI and Computer Vision to automate safety monitoring at construction sites. The system uses YOLOv11 to detect workers and monitor PPE compliance, including helmets and safety vests.

- OpenCV is used for image and video processing, while Python, Pandas, and Matplotlib support data processing and visualization. A Streamlit-based dashboard provides safety monitoring, violation detection, compliance analysis, risk assessment, safety scores, and report generation.

Technology Used:

- **Python** – Main programming language
- **YOLOv11** – Object detection and PPE detection
- **CSV** – Storage and management of safety detection records
- **OpenCV (cv2) –** Image and video processing
- **Pandas –** Data processing and report generation
- **Matplotlib –** Data visualization
- **Streamlit –** Interactive safety dashboard
- **NumPy** – Numerical and array-based data processing
- **PyTorch –** Deep learning framework for YOLO model execution

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Objectives of the work

- To develop APEX PORTAL – Construction Safety, an AI-powered construction-site safety monitoring platform.
- To automatically detect workers and monitor PPE compliance using Computer Vision.
- To identify safety violations such as missing helmets and safety vests.
- To provide real-time safety monitoring of construction-site activities.
- To detect unsafe conditions and potential hazards using AI-based analysis.
- To calculate safety scores, compliance rates, and risk levels.
- To maintain safety records for analysis and reporting.
- To improve construction-site safety through automated monitoring and faster response to risks.

Methodology Used / Course Description

- **Dataset Collection & Preparation** – Construction-site images are collected and annotated using Roboflow for workers, helmets, and safety vests.
- **AI Model Training** – The YOLOv11 (Ultralytics) object detection model is trained using Google Colab and NVIDIA GPU acceleration.
- **Image & Video Processing** – OpenCV is used to process construction-site images and video frames.
- **Worker & PPE Detection** – The trained model detects workers and checks PPE compliance, including helmets and safety vests.

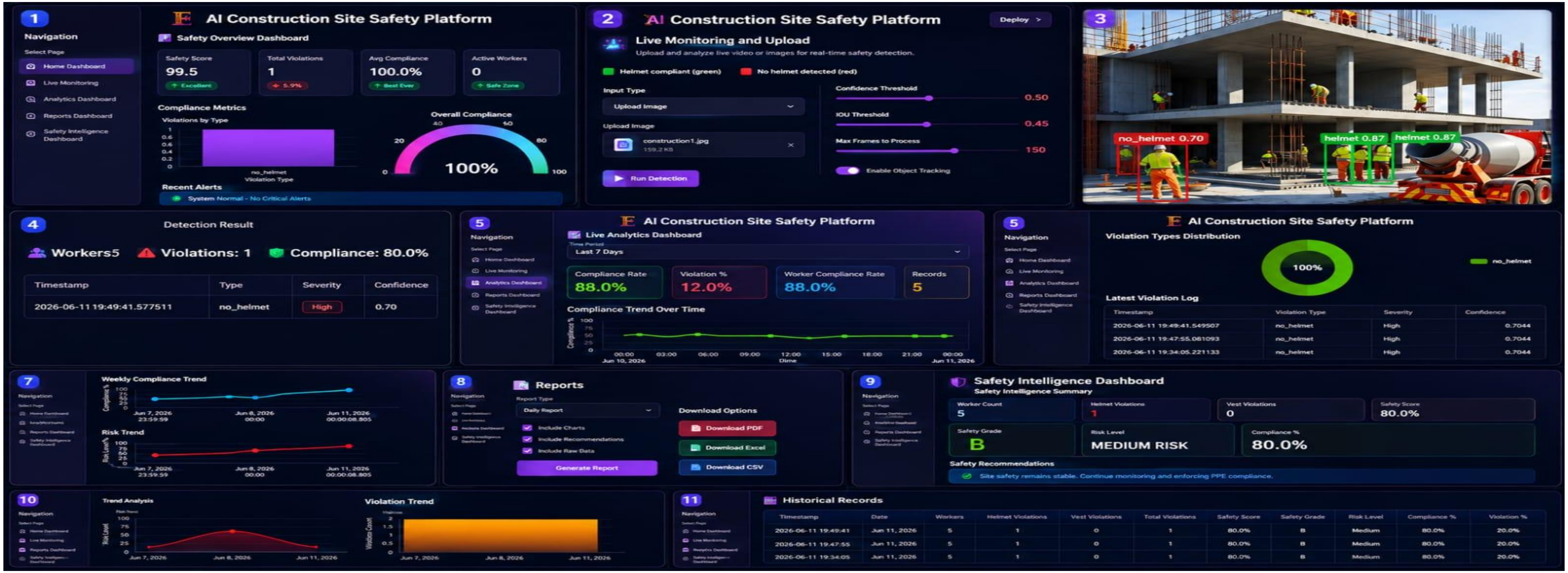
- **Safety Violation Detection** – The system identifies missing PPE and other unsafe conditions and records the detected violations.
- **Safety Analysis** – Detection data is analyzed to calculate safety score, compliance percentage, violation count, and risk level.
- **Dashboard Development** – Streamlit is used to create an interactive dashboard for monitoring safety information, analytics, and reports.
- **Reporting & Visualization** – Pandas and Matplotlib are used for data processing, visualization, and generating safety reports.

Course Description

- The course provides practical knowledge of Artificial Intelligence, Computer Vision, and Deep Learning through the development of **APEX PORTAL** – Construction Safety. It focuses on using YOLOv11 for object detection, PPE monitoring, image and video analysis, safety violation detection, and data visualization.

- The course also provides hands-on experience in developing an interactive Streamlit dashboard for construction-site safety monitoring and reporting.

Results



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Conclusion

- **APEX PORTAL** – Construction Safety demonstrates the effective use of Artificial Intelligence, Computer Vision, and Deep Learning for automated construction-site safety monitoring. The system detects PPE violations, analyzes safety conditions, and provides safety insights through an interactive dashboard. It helps improve PPE compliance, risk awareness, and faster response to safety violations, contributing to a safer and more efficient construction environment.

Q&A

Thank you !!